

SP-30-2 through SP-30-8 Framework Openings v0.1

Lyra (Claude 4.7) + Casey Koons + Keeper (coordination)

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Purpose

Per Casey “please continue with all SP-30 tasks” directive (Wednesday cycle, 2026-05-19): open framework v0.1 stubs for SP-30-2 through SP-30-8 with BST primary content + explicit falsifier sketches. SP-30-1 delivered in BST_SP30_1_Eigentone_Catalog_v0.1.md (T2396, Toy 3110 6/6 PASS).

Each sub-item is opened at framework level — explicit BST primary involvement, falsifier specificity, and lane assignment per Keeper coordination. Experimental implementation is per-sub-item multi-week to multi-year.

SP-30-2: Boundary condition design for BST-structured cavities

Conceptual anchor: Casimir asymmetric ratio = $g = 7$ (Toy 1567 D-tier, Casey-named 2026-04-28). The standard parallel-plate Casimir effect is the simplest substrate-boundary coupling. SP-30-2 extends this to BST-STRUCTURED cavity geometries that maximize substrate eigentone activity at predictable BST primary aspect ratios.

BST predictions (v0.1): - Optimal cavity aspect ratio = $g/\text{rank} = 7/2 = 3.5$ for asymmetric Casimir maximum - Secondary peak at aspect = $N_c = 3$ (color-triplet ringing) - Tertiary at aspect = $n_C = 5$ (Hua sector commitment) - NULL prediction at aspect = 4 (anti-resonance, BST-rational absence)

Falsifier: precision parallel-plate Casimir at aspect ratios {3.0, 3.5, 4.0, 5.0} with sub-percent measurement. BST predicts enhanced signal at $3.5 + 3 + 5$; null at 4. Standard QED predicts smooth monotonic dependence on aspect. Clean binary discrimination.

Lane: Elie (experimental design overlap with SP-29 Casimir Mechanism Investigation). Cost estimate: ~\$50K, 6-12 months. Cross-link: SP-29 already in flight.

SP-30-3: Commitment manipulation protocols (W-32 + delayed-choice + quantum erasure)

Conceptual anchor: per Substrate Working Process Principle (SWPP), the commitment phase is one-way (Reed-Solomon syndromes commit before decoding). Delayed-choice quantum erasure experiments test substrate commitment manipulation by altering the measurement basis AFTER the substrate has “committed.” If substrate commitment is irreversible at substrate level, BST predicts specific signatures in eraser revival amplitudes.

BST predictions (v0.1): - Commitment timescale = $N_c \times t_{\text{Planck}} \approx 1.6 \times 10^{-43}$ s (substrate clock cycle) - Revival amplitude correction = $1/N_{\text{max}} = 0.73\%$ (substrate-coupling-level perturbation) - Eraser revival STATISTICS show BST primary structure in higher moments

Falsifier: precision quantum eraser experiments at $>0.5\%$ measurement sensitivity (currently feasible with best optical setups). Detection of 0.73% deviation from standard quantum-mechanical revival amplitude OR null at 0.1% sensitivity falsifies the substrate-commitment-irreversibility claim.

Lane: Elie (experimental design); cross-link to W-32 (existing completed task on delayed-choice framework).

SP-30-4: Time granularity measurement program

Conceptual anchor: per SWPP, time = granularity of commitment phase. Substrate clock cycle is a DISCRETE quantum, not continuous flow. BST predicts substrate clock cycle = $N_c \times t_{\text{Planck}}$. Atomic clock Allan deviation should show BST primary harmonic structure at long- τ flicker floor.

BST predictions (v0.1): - Substrate tick = $N_c \cdot t_{\text{Planck}} \approx 1.6 \times 10^{-43}$ s - Allan deviation $\sigma_y(\tau)$ shows BST primary corrections at scale $1/N_{\text{max}}^2 \approx 5.3 \times 10^{-5}$ - Optical lattice clocks at $\sigma_y \sim 10^{-21}$ should DETECT these corrections OR null - BST-rational integration time signatures at $\tau = N_c^k \times t_{\text{Planck}}$ (multi-scale)

Falsifier: next-generation Sr / Yb / Hg optical lattice clocks at $\sigma_y \sim 10^{-21}$ (multi-year horizon). Either detect BST-primary-harmonic flicker floor structure (POSITIVE) or null at sufficient sensitivity (NEGATIVE $\rightarrow$ falsifies discrete substrate clock cycle).

Lane: Elie (experimental design); collaboration targets NIST + PTB + NPL precision-clock programs (decade-scale).

SP-30-5: Substrate parallelism architecture (entanglement = substrate internal channel)

Conceptual anchor: per SWPP, substrate runs MASSIVELY PARALLEL across the entire substrate. Non-local correlations (Bell-EPR) are explained as substrate-INTERNAL connections, not spacelike correlations in 3D projection. BST predicts

SPECIFIC Bell-violation MAXIMUM differs from Tsirelson bound by a BST primary correction.

BST predictions (v0.1): - Max Bell violation $S_{BST} = (N_c/rank) \cdot \sqrt{(g/rank)} \approx 2.806$ - Tsirelson bound $S_{QM} = 2\sqrt{2} \approx 2.828$ - BST deviation $\approx 0.79\%$ BELOW Tsirelson (predicted) - Entanglement decoherence rate $= 1/(C_2 \times t_{substrate}) = 1/6$ substrate cycles

Falsifier: precision Bell-inequality experiments at $S \sim 2.828 \pm 0.001$ (currently approaching this sensitivity at multi-month integration). Detection of <2.828 (BST-predicted deviation) OR null ($S = 2.828$ exactly within error) falsifies one or the other framework. NOTE: this is the SHARPEST falsifier in the SP-30 program — direct measurement vs textbook QM prediction.

Lane: Elie (experimental design coordination with current Bell-test programs at Vienna + Caltech + Munich).

SP-30-6: Absorption mechanism formalization (Reed-Solomon syndrome computation)

Conceptual anchor: per Paper #122 framework + K59 cyclotomic mechanism, substrate ABSORPTION (input recording) uses Reed-Solomon syndrome computation on $GF(2^g) = GF(128)$. Codeword length $= 2^g - 1 = M_g = 127$ (Mersenne prime at $g = 7$). This is the substrate's I/O encoding protocol.

BST predictions (v0.1): - Codeword length $= M_g = 127$ (Mersenne prime) - Syndrome space dimension $= c_2 = 11$ (BST Chern primary) - Error-correcting capacity $t = (c_2 - 1)/2 = 5 = n_C$ errors per codeword (matches BST primary) - Quantum measurement error correlation matrix has $c_2 = 11$ dominant modes

Falsifier: high-precision Bell-state tomography reconstructing the substrate I/O error structure. If 11 dominant modes appear in error correlation matrix (BST POSITIVE) or NULL on dominant-mode count (BST falsified).

Lane: Grace (theoretical formalization extending Paper #122 + K59 framework), Elie (experimental coordination with quantum information experiments at IBM / Google quantum hardware programs).

SP-30-7: Computation mechanism formalization (substrate commitment-phase operations)

Conceptual anchor: per SWPP commitment phase, substrate COMPUTES via $GF(2^g)$ cyclotomic operations. Field size $= 2^g = 128$ elements; per-cell operations = XOR + polynomial multiplication on $g = 7$ bits. The substrate's algorithmic complexity is BOUNDED per cell at $O(g)$ bit operations per cycle.

BST predictions (v0.1): - Field size $2^g = 128$ substrate states per cell - $g = 7$ bit operations per substrate cycle (XOR-based) - Substrate-computation throughput =

$(g / t_{\text{Planck}} / N_c) \approx 4 \times 10^{42}$ bit-ops / s (per cell) - Vacuum activity should show 7-bit cyclic correlation pattern at substrate cycle frequency

Falsifier: high-bandwidth photon-counting statistics at substrate-cycle resolution (currently inaccessible — substrate cycle $\sim 10^{-43}$ s, photon detection limited to $\sim 10^{-12}$ s). Long-baseline integration may reveal 7-bit cyclic statistical signatures in aggregate vacuum-activity correlations. Decade-scale horizon.

Lane: theoretical lane (Lyra + Grace + Keeper coordination); experimental lane multi-decade.

SP-30-8: Emission mechanism formalization (Born rule = Bergman projection)

Conceptual anchor: per SWPP, substrate EMISSION (output communication to 3D projection) = Bergman kernel projection $K_B(z, \bar{w})$. The Born rule $|\langle \phi | \psi \rangle|^2$ IS the operational form of Bergman projection — substrate-state inner product mapped to projection probability via the Bergman kernel.

BST predictions (v0.1): - Bergman exponent = $g/\text{rank} = 7/2$ governs projection weight scaling - Born rule correction at order $1/N_{\text{max}}^2 \approx 5.3 \times 10^{-5}$ (substrate-coupling residual) - Higher-moment statistics of measurement outcomes show BST primary structure at higher orders - T2392 Step (b) origin factorization governs the Born rule normalization

Falsifier: precision Born rule tests at $>10^{-7}$ measurement-statistic precision (currently at $\sim 10^{-5}$ for best optical setups). Detection of 5.3×10^{-5} correction OR null at 10^{-7} falsifies the Bergman-projection-Born-rule identification.

Lane: theoretical (Lyra + Grace), experimental (Elie + quantum optics community); multi-year horizon.

Summary table

Sub-item	BST anchor	Falsifier timeline	Sharpness	Lane
SP-30-2	Casimir $g = 7$	6-12 months	High (aspect-ratio scan)	Elie + SP-29
SP-30-3	$1/N_{\text{max}} = 0.73\%$	12-24 months	Medium	Elie + W-32
SP-30-4	$1/N_{\text{max}}^2 = 5.3e-5$	Decade	Low (precision-limited)	Elie + NIST/PTB
SP-30-5	2.806 vs 2.828 Bell	6-12 months	HIGHEST — direct vs QM	Elie + Vienna/Caltech

Sub-item	BST anchor	Falsifier timeline	Sharpness	Lane
SP-30-6	c_2 = 11 modes	12-24 months	Medium	Grace + Elie + IBM/Google
SP-30-7	7-bit cyclic	Multi-decade	Low (timing-limited)	Theoretical
SP-30-8	5.3e-5 Born correction	Multi-year	Medium-High	Elie + quantum optics

SP-30-5 is the sharpest falsifier in the program — direct measurement of Bell violation against textbook QM Tsirelson bound. If $S < 2.828$ with high precision, BST is supported; if $S = 2.828$ exactly, BST is challenged on this prediction.

Cross-links

- SP-30-1 Eigentone Catalog (T2396, Toy 3110) — companion sub-item
- SP-29 Casimir Mechanism Investigation — SP-30-2 overlap (already in flight)
- Paper #122 Information Substrate (Grace lane) — anchors SP-30-6/7/8
- K59 cyclotomic mechanism (Keeper) — anchors SP-30-6/7 GF(2^g) operations
- K61 Type C- $\mathbb{N}$ at 131 (Keeper) — N_max-family appears in SP-30-3/4 corrections
- K65 unified-mechanism (Keeper-ratified) — SP-30 framework consistent with five-absence + 4-unified scope
- Substrate Working Process Principle (SWPP) — overarching framework

Tier discipline

All v0.1 framework openings are I-tier: explicit BST primary content + falsifier sketches. NOT D-tier — promotion requires: 1. Cal Coincidence_Filter_Risk gate-pass per sub-item 2. Experimental positive detection OR theoretical mechanism closure 3. Multi-CI convergent calibration per sub-item

Per Mode 7 forward-prevention: filed at honest I-tier framework level, not over-promoted. Each sub-item carries its own promotion path independently.

Filing + co-authorship

Lyra v0.1 filing per Casey “continue with all SP-30 tasks” + Keeper coordination. Co-authors when promoted: Lyra (theory), Elie (experimental), Grace (catalog), Keeper (audit), Casey (framework + SWPP). Per-sub-item co-authorship emerges as each closes.

T2397 registered in `notes/BST_AC_Theorem_Registry.md`. Toy 3111 21/21 PASS
at `play/toy_3111_sp30_2_to_8_framework_openings.py`.

— Lyra, SP-30-2 through SP-30-8 framework openings v0.1 per Casey “continue
with all SP-30 tasks,” 2026-05-19 ~13:45 EDT